EE561 Advanced Static Power Conversion · Multilevel Converters module

Potential Final Questions: Modulation and Balancing

From the joint proposal by Sefer İdacı and M. Aytunç Evlice, for M. Aytunç Evlice's talk

Two questions for a closed-book exam with the instructor's cheat sheet, each designed for about 20 minutes and 25 points, on modulation and balancing: three-level space-vector modulation and a neutral-point balancing loop. Every question asks for a derivation, a numerical result and an engineering interpretation; full solutions and a marking guide follow on the later pages. All numbers differ from the lecture examples.

Question 1 · Three-level space-vector modulation

MODULATION — **25 pts**

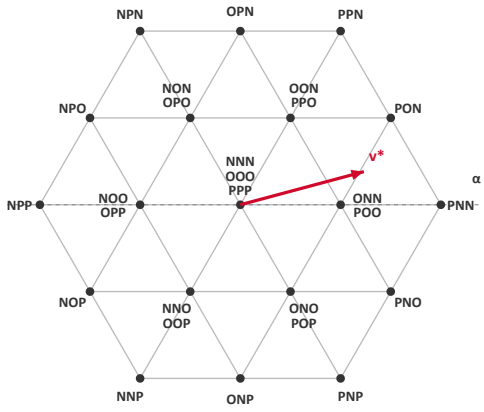
The figure shows the space-vector diagram of a 3-level NPC converter; vectors are in units of $V_{dc}/2$ (small vectors 2/3, medium 2/√3, large 4/3). The reference is $\mathbf{v}^* = 0.85 \angle 15°$.

a. Identify the three nearest vectors (their switching states) and give their αβ coordinates. [4]

b. Using volt-second balance, compute the dwell-time fractions $d_k = T_k/T_s$. [8]

c. Construct a symmetric seven-segment sequence in which exactly one phase changes by one level at each transition. Give each segment's duration as a fraction of T_s (split the redundant small vector equally). [6]

d. Give the midpoint current i_O of every state in your sequence in terms of the phase currents. If $\Delta V = v_{C1} - v_{C2} > 0$ and $i_a > 0$, how should the small-vector time be redistributed (with $C\,d\Delta V/dt = i_O$, i_O drawn from O)? Does this change the output voltage? [7]



3-level space-vector diagram, states as phase levels (a b c).

Question 2 · Designing a neutral-point balancing loop

BALANCING AND CONTROL — **25 pts**

A 3-level NPC with carrier PWM has a split DC link of 2 × 3.3 mF; an ideal source holds $v_{C1} + v_{C2}$ constant, so $C\,d\Delta V/dt = i_O$ with $\Delta V = v_{C1} - v_{C2}$ and i_O drawn from the midpoint. The phase currents are balanced with $\hat{I} = 40$ A, and $m = 0.9$. A common offset r_0 is added to all references.

a. Starting from $\langle i_O \rangle = \Sigma_x(1 - |r_x + r_0|)\, i_x$, derive $\partial\langle i_O\rangle/\partial r_0$ and its average over a fundamental period for $r_x = m \cos\theta_x$, $i_x = \hat{I}\cos(\theta_x - \phi)$. [6]

b. With $r_0 = K \cdot \mathrm{sign}(P) \cdot \Delta V$, derive the closed-loop time constant and choose K for $\tau = 20$ ms at $\cos\phi = 0.8$. [6]

c. With the same K, what is τ at $\cos\phi = 0.2$? What happens in rectifier mode if sign(P) is omitted? [5]

d. At unity power factor the midpoint current contains a third harmonic of amplitude $8m\hat{I}/(5\pi)$. Estimate the peak-to-peak ripple of ΔV. Can — and should — the balancing loop remove this ripple? [8]

Solutions and marking guide

Solution 1 · Three-level space-vector modulation — 25 pts

(a) $\mathbf{v}^* = 0.85\angle 15° = (0.821, 0.220)$. Nearest vectors: small **ONN/POO** (0.667, 0); large **PNN** (1.333, 0); medium **PON** (1.000, 0.577). [4]

(b) β: $0.577\, d_{PON} = 0.220 \rightarrow d_{PON} = 0.381$; $d_S + d_L = 0.619$. α: $0.667\, d_S + 1.333\, d_L + 0.381 = 0.821 \rightarrow d_L = 0.041$, $d_S = 0.578$. [8]

(c) ONN (0.145) → PNN (0.020) → PON (0.190) → POO (0.289) → PON (0.190) → PNN (0.020) → ONN (0.145). Transitions: phase a O→P, phase b N→O, phase c N→O, then back in reverse; each phase switches once up and once down per period. [6]

(d) i_O: ONN = $+i_a$; PNN = 0; PON = $+i_b$; POO = $i_b + i_c = -i_a$. To reduce ΔV we need $\langle i_O \rangle < 0$: **lengthen POO and shorten ONN**, keeping their sum at 0.578. Both states produce the same space vector, so the output (line-to-line) voltage does not change — only the midpoint charge does. [7]

Solution 2 · Neutral-point balancing loop — 25 pts

(a) For an offset small enough not to change the sign of any r_x: $\partial\langle i_O\rangle/\partial r_0 = -\Sigma_x \operatorname{sign}(r_x)\, i_x$. Since $\langle \operatorname{sign}(\cos\theta)\cos(\theta-\phi)\rangle = (2/\pi)\cos\phi$ for each phase, the period average is $\mathbf{-(6/\pi)\,\hat{I}\cos\phi}$. [6]

(b) $C\, d\Delta V/dt = -(6/\pi)\hat{I}\cos\phi \cdot K \operatorname{sign}(P)\, \Delta V = -(6/\pi)\hat{I}|\cos\phi|\, K\, \Delta V \rightarrow \mathbf{\tau = \pi C/(6\hat{I}K|\cos\phi|)}$. For $\tau = 20$ ms: $K = \pi \cdot 3.3\times10^{-3}/(6 \cdot 40 \cdot 0.8 \cdot 0.02) = \mathbf{2.7 \times 10^{-3}\ V^{-1}}$ (an offset of 0.027 per 10 V of imbalance). [6]

(c) $\tau \propto 1/|\cos\phi|$: $\tau = 20\text{ ms} \times 0.8/0.2 = \mathbf{80\ ms}$. Without sign(P), $\cos\phi < 0$ in rectifier mode makes the loop gain positive: ΔV grows exponentially until saturation and an overvoltage trip. [5]

(d) $\hat{I}_{O3} = 8 \cdot 0.9 \cdot 40/(5\pi) = 18.3$ A at 150 Hz. Integrating, $\Delta V_{pp} \approx 2\hat{I}_{O3}/(3\omega C) = 2 \cdot 18.3/(3 \cdot 314 \cdot 3.3\times10^{-3}) \approx \mathbf{11.8\ V}$ (a numerical integration gives 11.9 V). The loop bandwidth is about $1/(2\pi\tau) \approx 8$ Hz, far below 150 Hz, so it cannot remove the ripple — and it should not try: the ripple is inherent to carrier PWM with zero average, and chasing it would only inject large offsets. It is handled by sizing C (or by modified modulation). [8]